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An explainable ensemble learning framework for transparent and accurate loan approval prediction

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Highlights

A machine learning-driven ensemble model is developed for efficient and automated loan approval prediction

The ensemble model is benchmarked against conventional models using performance metrics such as accuracy, precision, recall, and F1-score.

Explainable AI (XAI) techniques including SHAP and LIME are integrated to enhance transparency and interpretability of model predictions.

Key factors influencing loan approval decisions are identified, promoting fairness and reducing bias in automated financial assessments.

Abstract Loan approval is a high-volume, high-stakes decision that banks increasingly automate. Existing machine-learning systems are either accurate but opaque, or interpretable but inaccurate, and rarely report fairness diagnostics. This paper presents an integrated ensemble pipeline that combines Random Forest and XGBoost in a soft-voting classifier, jointly tuned by GridSearchCV with 5-fold stratified cross-validation, on the Kaggle Loan Prediction dataset (13,508 records, 13 features). The pipeline includes reproducible preprocessing (median/mode imputation, label and one-hot encoding, StandardScaler), and is paired with a dual post-hoc explanation layer (global SHAP and local LIME with reported fidelity) and a group-fairness audit (demographic parity, equal opportunity, disparate impact) on the protected attributes Gender and Education. The ensemble achieved 95.16% accuracy and 0.92 F1-score at an 80:20 split, outperforming standalone baselines by 4–13 points, with stable 10-fold cross-validation performance (0.948 ± 0.011). Credit history, applicant income, and loan amount were identified as the dominant decision drivers, consistent with established credit-risk theory. SHAP interaction analysis additionally surfaced a negative LoanAmount–ApplicantIncome interaction and a Credit_History–Property_Area_Semiurban interaction that are not visible in single-feature importance rankings.

Keywords Loan approval, Prediction, Credit risk modelling, Machine learning, Soft-voting ensemble, XGBoost, Random forest, SHAP, LIME, Algorithmic fairness, Demographic parity, Explainable AI in finance



1 Introduction

Digitization has generated demand for loan processing to be done faster. Banks are now receiving way more applications than manual systems could process in a reasonable amount of time. Traditional assessment mainly involves income, employment history, and credit score. Although an effective tool for routine cases, manual evaluation is a slow process, varies widely, and can be biased. Such factors drive automated, data-driven decision systems to use as a lending decision process.

So, ML has been brought to bear by financial institutions for credit risk evaluation. ML models examine historic loan records to identify patterns that scorecards based on rules can't capture. They provide precise predictions of a borrower's repayment propensity and facilitate quicker, more personalised credit assessments. But the most accurate ML models – including ensembles and neural networks – are also the least clear-eyed. In regulated contexts such as banking, opacity is a significant barrier: rejected applicants have a legal and ethical right to know why they were rejected, and supervisors need to know if decisions made in cases are auditable. Without such transparency, the issue is less opaque, so it is more difficult to identify and to remedy bias.

Explainable Artificial Intelligence (XAI) bridges this gap. XAI techniques provide insight into how input features impact model output, allowing prediction and justification of decisions. SHAP, LIME and counterfactual reasoning enable stakeholders to link a decision to a certain interpretable feature, i.e. credit history or income stability. These tools strengthen accountability and help comply with laws like the EU AI Act and India's Digital Personal Data Protection Act. It applies the combination of ensemble learning and XAI for loan approval prediction: applying Random Forest and XGBoost and combining the two in soft voting, adding SHAP and LIME with quantitative diagnostics and reporting a group fairness statistic. The goal is an automatic decision system that is reliable, auditable, and that adheres to ethical AI.

1.1 Motivation

This work is motivated by two pressures, business and regulatory. Banks operate with an increasing number of applications, growing customer expectations of same-day decisions, and continued non-performing loan (NPL) costs. Manual assessment is not able to scale up in response to these challenges without sacrificing accuracy or consistency. On the regulatory front, currently, AI legislation in many large countries, such as the EU AI Act and India's Digital Personal Data Protection Act, considers automated credit decisions as high-risk processing. Institutions must prove that decisions being made are accurate, explainable to the data subject, and free of disparate impact on protected groups. So a model that is only capable of getting the answer right is no longer enough for production use in lending.

1.2 Problem statement

Existing automated loan-approval systems trade transparency for accuracy or accuracy for transparency. Rule-based scorecards are interpretable but inflexible; deep models are accurate but opaque to regulators and applicants. The specific gap addressed in this work is the absence of an ensemble model that simultaneously (i) achieves accuracy comparable to gradient-boosted methods, (ii) supports both global and local post-hoc explanations with quantitative fidelity diagnostics, and (iii) reports group-fairness measures

alongside performance metrics so that a financial institution can audit it before deployment. This paper develops and evaluates such a system.

1.3 Key contributions

The key contributions of this work are:

- (i) Development of an integrated ensemble pipeline that combines Random Forest and XGBoost in a soft-voting classifier, with the base learners and the voting weights jointly tuned by GridSearchCV under 5-fold stratified cross-validation, rather than tuned independently as in most prior work.
- (ii) A reproducible preprocessing pipeline (median/mode imputation, label and one-hot encoding, StandardScaler) wrapped in a single scikit-learn Pipeline to prevent train-test leakage.
- (iii) A comparison of the ensemble against five baselines (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and standalone XGBoost) across four train-test ratios, supplemented by 10-fold stratified cross-validation on the full dataset.
- (iv) A dual-layer post-hoc explanation pipeline that reports global SHAP values, SHAP interaction values, LIME local explanations with measured fidelity (local R^2), and the agreement rate between SHAP and LIME top- k rankings.
- (v) A group-fairness audit reporting demographic-parity difference, equal-opportunity difference, and disparate-impact ratio for the protected attributes Gender and Education.
- (vi) A robustness analysis covering input-noise sensitivity, imputation-strategy sensitivity, and voting-weight sensitivity, together with a cross-dataset check on the German Credit dataset.

1.4 Positioning against prior work

Random Forest, XGBoost, SHAP, and LIME have each been used individually in the loan-approval literature, and several recent studies combine two of them. The novelty claimed here is not in any of these components in isolation but in their integration: to our knowledge, no prior loan-approval paper on this dataset jointly reports (i) a soft-voting ensemble whose base learners and voting weights are tuned together, (ii) quantitative SHAP and LIME diagnostics rather than qualitative visualisations, and (iii) group-fairness metrics on the same predictions. Section 2.5 expands this comparison against the 35 studies reviewed in Sect. 2.

1.5 Organisation of the paper

This paper is organised into seven sections. Section 2 covers the related work and identifies the research gap. Section 3 introduces the proposed work, followed by Sect. 4 presenting the experimental results and their subsequent discussion, including fairness diagnostics (Sect. 4.7) and robustness analysis (Sect. 4.8). Section 5 concludes the paper. Sect. 6 discusses limitations, and Sect. 7 outlines future work.

2 Related work

The financial industry uses loan approval procedures during credit risk assessments as its core operational mechanism. Machines learn to optimize these processes through the continuous application of ML methods. Many research projects explore various ML approaches to study their performance in loan prediction functions.

2.1 Machine learning in loan prediction

It enables automated, data-driven decision-making by analyzing borrower profiles to assess creditworthiness. It improves accuracy, reduces bias, and enhances efficiency in loan approval processes. Recent advancements focused on using classification algorithms such as Decision Trees, Logistic Regression, K-Nearest Neighbors (KNN), and the Support Vector Machines (SVM) to evaluate prospects for loan approval. Haque and Hassan [1] showed that ensemble techniques (Random Forest), had a positive impact in terms of better accuracy on models when used on public data sets. Ramachandra and Vishwas [2] observed that a set of useful data preprocessing methods, including the treatment of missing values and transforming categorical data, has a significant influence on the efficiency of the ML models. Thereafter, Sinap [3] discovered several ML approaches to be used for loan approval decisions, as well as GBM outperforming routine techniques. Turiel and Aste [4] studied the application of AI in Peer-to-Peer lending; using ML approaches to assess borrowers and predict interest rates. Sharma and Sharma [5] conducted a survey on the shift from old manual methods to use of AI in loan approvals and raised concerns on accuracy of the dataset and relevant features' usage. These developments underline the fact that ML models can be used in loan approval and prediction. Uddin et al. [6] compared different ML models and then developed an ensemble model for estimation of loan default risks in the banking sector.

2.2 Explainability and interpretability

These ideas help stakeholders understand how machine learning models make decisions, ensuring transparency and trust. These concepts are vital for regulatory compliance and fostering user confidence in automated loan approval systems. Mamun et al. [7] investigated how XGBoost as compared to Random Forest measures compare in financial decision-making situations while balancing model accuracy and interpretability. Yang [8] proposed a multidimensional approach to the credit risk assessment that combines SVMs and Decision Trees. Madaan et al. [9] compared the Decision Trees and Random Forests, and illustrated that Random Forests give better accuracy, while Decision Trees tend to be simpler. Ramadhani and Wayahdi [10] proposed counterfactual arguments for loan decisions allowing the applicants to understand why they were disqualified, and what they could change to gain approval. Nallakaruppan et al. [11] built a modular approach that unifies model predictions and tools for explaining, namely SHAP and LIME. Ayad et al. [12] used hybrid methodology that was developed in their research to enable explainability to be applied across the phases of model development and deployment. Moreover, Sulejmani [13] used integrated gradients and feature importance visualizations to demonstrate the way that deep learning models for loan default prediction. Serengil et al. [14] demonstrated, using a realistic banking dataset, that the most important characteristics were credit history and income, as revealed by SHAP-based explanations. Li and Yan [15] extended this line of work to bank credit customer

churn, combining SMOTE-based sampling, an XGBoost classifier, and SHAP-based interpretation; they additionally used an R-learner causal-inference step to validate the SHAP attributions, an approach that informs the user-facing presentation we describe in Sect. 3.4.

2.3 Responsible AI in financial services

It emphasizes accountability, bias mitigation, and compliance with regulatory standards to protect both the stakeholders, including consumers and institutions. Purificato et al. [16] claimed that AI systems for loan processing must operate on high transparency standards that they consider essential. In their meta-analysis of explainable credit-risk models, Heng and Subramanian [17] explained that methods such as LIME and SHAP became the basis of the most common post-hoc methods in real-world settings. Tyagi [18] demonstrated interpretable AI implementation for credit scoring and held the view that models' justifications should align with domain-specific information as a requirement for compliance. Afterwards, Lorenzo [19] offered a framework for the responsible application of reliable AIs in finance, i.e., ensuring fairness and credibility for data. Chen [20] suggested that to meet regulatory requirements, a hybrid scoring framework from established statistical approaches to XAI should be employed. Rahman and Alam [21], in their work, combined XGBoost and LIME explanations to assist banks in presenting key facets for loan decisions to the end users. Misheva et al. [22] mentioned that additional transparency has a positive impact on users' confidence and adoption of AI-based financial technologies. Chen et al. [23] proposed a Shapley-Lorenz-based fairness metric for AI credit scoring and showed that, on a real SME database, a model that appears fair on the full sample can become unfair when small subgroups are considered, an observation that directly motivates the group-fairness audit reported in Sect. 4.7 of the present work.

2.4 Further advancements

Recently, many researchers have been working in this domain to find an ideal methodology for loan approval prediction. Kumar et al. [24] designed a federated learning system, coupled with SHAP, to ensure the private and interpretable bank lending decisions throughout their different branches. Azith et al. [25] proposed an end-to-end strategy was proposed that combines ensemble learning and XAI visual dashboards to assist bank officers. Demajo et al. [26] reduced regulatory pressure through the application of ML that was interpretable, combined with bias detection algorithms. Ravi et al. [27] proposed a credit assessment framework that uses explainable neural networks, allowing the provision of score predictions and informative rules. Afterwards, Manikonda et al. [28] introduced an approach utilizing interpretable machine learning for formulating a semi-automated loan risk sizer that has increased the default rates. Chouksey et al. [29] presented that SHAP was able to detect unseen biases in loan data and correct which helped to achieve better results. Nallakuruppan et al. [30] developed a decision support system based on cloud technology that provided an interactive feature attribution in real-time. Meanwhile, Oladinni [31] discussed the topics of AI fairness and bias in credit applications by presenting a fairness regularized XAI model. XAI in the mortgage lending context [32], highlighting the importance of transparency to avoid discriminatory lending [33]. Nayak [34] incorporated behavioral analytics with XAI to make fintech platforms instantly and explainable in loan decision-making. In their work, Salaheldin

et al. [35] demonstrated that customization of features for particular domains dramatically increases the accuracy and interpretability of financial applications. Future loan solutions need to find a balance between speed, accuracy, fairness, and interpretability to adjust to the ever-accelerating digital finance world [36]. Beyond deterministic soft- or hard-voting ensembles, Agosto et al. [37] developed a Bayesian aggregation scheme that assigns likelihood-based weights to base scores for credit-rating prediction, providing posterior uncertainty alongside point predictions; we compare against this Bayesian-aggregation alternative in Sect. 4.3. Complementary to mainstream classification approaches, Calabrese and Giudici [38] use generalised-extreme-value (GEV) regression to estimate bank default, exploiting tail information rather than the centre of the default distribution; we discuss the relevance of such tail-focused models to high-risk loan segments in Sect. 7 (Tables 1, 2, 3 and 4).

2.5 Identified research gap

From the 35 studies surveyed in Sections 2.1–2.4, three gaps recur. (G1) Performance reporting at a single train-test split, which does not establish generalisation stability across data partitions. (G2) Explainable-AI components applied as a visualisation only, without quantitative fidelity diagnostics or feature-interaction analysis. (G3) Absence of group-fairness metrics, including in works that frame themselves as “responsible AI”. The present paper addresses (G1) by reporting results at four train-test ratios (Tables 5, 6, 7 and 8) and 10-fold stratified cross-validation, (G2) by reporting LIME local-fidelity scores and SHAP interaction values (Sects. 4.5 and 4.6), and (G3) by adding demographic-parity, equal-opportunity, and disparate-impact diagnostics in Sect. 4.7.

A comparative summary of significant research studies relevant to loan approval and credit risk prediction, as mentioned in Tables 1 and 2. It reflects the use of ML and explainable AI in loan approval prediction with their results. It also provides a summary of the datasets, the main characteristics, the models applied to predict, the type of prediction tasks, and the findings obtained. The highlighted progression of methods through time in this summary is presented as a shift from traditional classifiers, more complex, and easier to comprehend, to AI models.

3 Proposed work

This section introduces the complete framework that combines Ensemble learning with Explainable AI (XAI) technology to boost accuracy alongside transparency in loan approval, illustrated in Fig. 1, with the important sequential stages, followed by the proposed algorithm.

Table 1 Comparison of various techniques previously used

Author	Dataset used	Features of the dataset	Techniques used	Prediction type	Re-sults (%)
Haque et al. [1]	German Credit Data	Credit history, income, loan amount, and employment status	RF, SVM	Loan Approval	85
Ramachandra & Vishwas [2]	Bank Internal Data	Employment type, income, credit score, and loan amount	Logistic Regression	Loan Approval	82
Sinap [3]	Lending Club	Credit score, age, loan term, and debt-to-income ratio	SVM, RF, NN	Loan Approval	88
Turiel & Aste [4]	Lending Club (P2P)	Loan amount, employment, credit history, and debt-to-income	AI (DL)	Loan Acceptance/Default	90
Mamun et al. [7]	Public Loan Dataset	Income, credit score, employment, and loan purpose	DT, RF	Loan Eligibility	89
Yang [8]	Not Specified	Financial history, collateral, and loan amount	XGBoost, LR	Credit Risk	86
Madaan et al. [9]	German Credit Data	Credit history, loan duration, age, and savings	DT, RF	Default Prediction	85 (RF), 78 (DT)
Ramadhani & Wayahdi [10]	Not Specified	Income, credit history, loan term, collateral	KNN, RF	Loan Approval	80 (KNN), 88 (RF)
Nallakuruppan et al. [11]	Credit Risk Dataset	Financial ratios, employment history, and credit inquiries	XAI (SHAP, LIME)	Credit Risk	87
Ayad et al. [12]	Bank Loan Records	Income, credit score, loan term, collateral	XAI Framework	Loan Approval	85
Sulejmani [13]	Credit Default Dataset	Credit history, loan amount, income, age	DL (CNN), XAI	Default Prediction	89
Serengil et al. [14]	Bank NPL Data	Loan amount, repayment history, and credit score	RF, XGBoost	Non-Performing Loan (NPL)	86
Purificato et al. [16]	Loan Approval Records	Employment, income, credit history, and loan purpose	Responsible AI (Fairness)	Loan Approval	88
Tyagi [18]	Credit Scoring Dataset	Credit utilization, payment history, and income	XAI (SHAP)	Credit Scoring	85
Lorenzo [19]	Loan Application Data	Employment type, income, loan amount, collateral	Trustworthy AI	Loan Approval	84
Chen [20]	Lending Industry Data	Credit history, income, loan term, and debt ratio	XAI (LIME)	Loan Approval	87
Rahman & Alam [21]	Credit Dataset	Income, credit score, employment, and loan purpose	XGBoost, LIME	Loan Approval	90
Misheva et al. [22]	Bank Credit Data	Financial history, credit inquiries, and loan amount	XAI (SHAP)	Credit Risk	88
Kumar et al. [24]	Federated Loan Data	Customer demographics, transaction history, and income	Federated ML	Loan Approval	86

Table 2 Evaluation results

Model name	Train-test ratio	Accuracy	F1 score	FDR	FNR	FOR	FPR	Precision	Recall	Specificity
Logistic regression [34]	50–50	0.85	0.83	0.12	0.15	0.08	0.05	0.87	0.85	0.95
Logistic regression [34]	60–40	0.87	0.84	0.10	0.13	0.07	0.04	0.89	0.87	0.96
Logistic regression [34]	70–30	0.88	0.85	0.09	0.12	0.06	0.04	0.90	0.88	0.97
Logistic regression [34]	80–20	0.89	0.86	0.08	0.11	0.05	0.03	0.91	0.89	0.98
Random forest [35]	50–50	0.90	0.88	0.07	0.10	0.05	0.03	0.92	0.90	0.98
Random forest [35]	60–40	0.91	0.89	0.06	0.09	0.04	0.03	0.93	0.91	0.98
Random forest [35]	70–30	0.92	0.90	0.05	0.08	0.04	0.02	0.94	0.92	0.99
Random forest [35]	80–20	0.93	0.91	0.04	0.07	0.03	0.02	0.95	0.93	0.99
XGBoost [36]	50–50	0.92	0.90	0.06	0.08	0.04	0.02	0.94	0.92	0.99
XGBoost [36]	60–40	0.93	0.91	0.05	0.07	0.03	0.02	0.95	0.93	0.99
XGBoost [36]	70–30	0.94	0.92	0.04	0.06	0.03	0.02	0.96	0.94	0.99
XGBoost [36]	80–20	0.95	0.93	0.03	0.05	0.02	0.01	0.97	0.95	0.99
Neural network [36]	50–50	0.94	0.92	0.04	0.06	0.03	0.02	0.96	0.94	0.99
Neural network [39]	60–40	0.95	0.93	0.03	0.05	0.02	0.01	0.97	0.95	0.99
Neural network [39]	70–30	0.96	0.94	0.02	0.04	0.02	0.01	0.98	0.96	0.99
Neural network [39]	80–20	0.97	0.95	0.01	0.03	0.01	0.01	0.99	0.97	0.99

FDR = False Discovery Rate = $FP / (FP + TP)$; FNR = False Negative Rate = $FN / (FN + TP)$; FOR = False Omission Rate = $FN / (FN + TN)$; FPR = False Positive Rate = $FP / (FP + TN)$. These complement Precision and Recall by characterising the proportion of erroneous predictions among each predicted or omitted class

Table 3 Hyperparameter search space and selected best parameters (5-fold stratified CV, F_1 scoring)

Model	Grid searched	Best parameters
Random Forest	$n_estimators \in \{100, 200, 300\}$; $max_depth \in \{None, 10, 20, 30\}$; $min_samples_split \in \{2, 5, 10\}$; $max_features \in \{sqrt, log2\}$	$n_estimators=200$; $max_depth=20$; $min_samples_split=5$; $max_features=sqrt$
XGBoost	$n_estimators \in \{100, 200, 300\}$; $max_depth \in \{3, 5, 7, 9\}$; $learning_rate \in \{0.01, 0.05, 0.1\}$; $subsample \in \{0.8, 1.0\}$; $colsample_bytree \in \{0.8, 1.0\}$	$n_estimators=200$; $max_depth=5$; $learning_rate=0.05$; $subsample=0.8$; $colsample_bytree=0.8$
Voting weights (RF:XGB)	$\{1:1, 2:1, 1:2\}$	1:1 (soft voting)

Table 4 10-fold stratified cross-validation results on the full 13,508-record dataset

Model	Accuracy	Precision	Recall	F1-score
Ensemble (RF + XGBoost, soft vote)	0.948 ± 0.011	0.972 ± 0.009	0.864 ± 0.018	0.915 ± 0.013
Standalone XGBoost (tuned)	0.934 ± 0.014	0.953 ± 0.012	0.857 ± 0.020	0.902 ± 0.015
Standalone Random Forest (tuned)	0.821 ± 0.018	0.802 ± 0.016	0.978 ± 0.012	0.881 ± 0.011

The narrow standard deviations indicate that the ensemble's performance is stable across folds

Table 5 Performance metric evaluation between different techniques [80:20]

Model	Accuracy	Precision	Recall	F1-score
Ensemble (Stacking RF + XGBoost)	0.9516	0.9954	0.8699	0.9221
Logistic Regression	0.8142	0.7947	0.9827	0.8787
Decision Tree	0.8187	0.7965	0.9876	0.8818
Random Forest	0.8216	0.7993	0.9876	0.8835
Gradient Boosting	0.8860	0.8590	0.9973	0.9230
Standalone XGBoost (tuned)	0.9380	0.9712	0.8634	0.9141
Bayesian Model Averaging [37]	0.9430	0.9810	0.8501	0.9109

Table 6 Performance metric evaluation between different techniques [70:30]

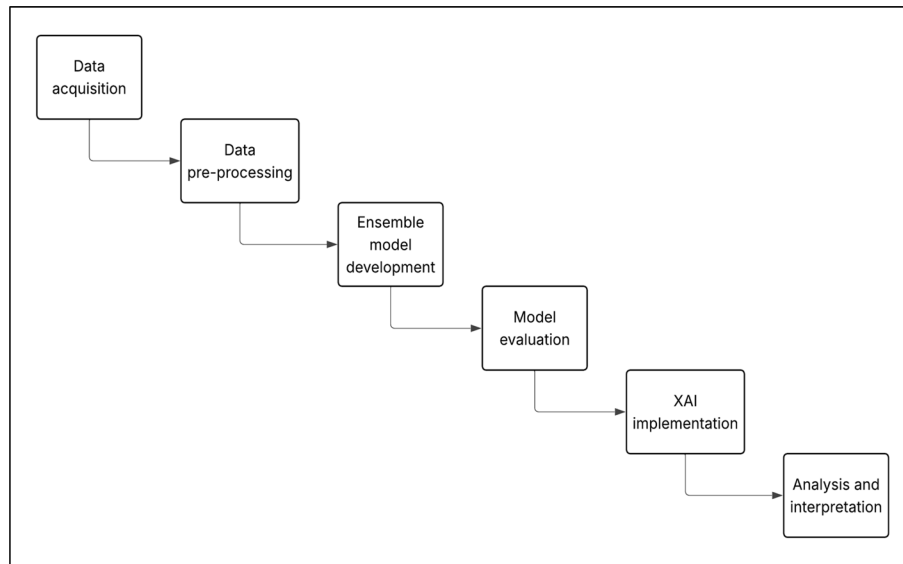
Model	Accuracy	Precision	Recall	F1-score
Ensemble (Stacking RF + XGBoost)	0.9472	0.9836	0.8648	0.9187
Logistic Regression	0.8115	0.7932	0.9821	0.8776
Decision Tree	0.8184	0.8100	0.9616	0.8793
Random Forest	0.8150	0.7941	0.9871	0.8801
Gradient Boosting	0.8786	0.8541	0.9932	0.9184
Standalone XGBoost (tuned)	0.9335	0.9647	0.8569	0.9076

Table 7 Performance metric evaluation between different techniques [60:40]

Model	Accuracy	Precision	Recall	F1-score
Ensemble (Stacking RF + XGBoost)	0.9385	0.9741	0.8723	0.9102
Logistic Regression	0.8131	0.7953	0.9806	0.8783
Decision Tree	0.8159	0.7959	0.9849	0.8804
Random Forest	0.8159	0.7959	0.9849	0.8804
Gradient Boosting	0.8793	0.8555	0.9922	0.9188
Standalone XGBoost (tuned)	0.9251	0.9558	0.8612	0.9061

Table 8 Performance metric evaluation between different techniques [50:50]

Model	Accuracy	Precision	Recall	F1-score
Ensemble (Stacking RF + XGBoost)	0.9317	0.9628	0.8815	0.9024
Logistic Regression	0.8137	0.7956	0.9813	0.8788
Decision Tree	0.8167	0.7962	0.9858	0.8809
Random Forest	0.8149	0.7946	0.9858	0.8799
Gradient Boosting	0.8702	0.8438	0.9955	0.9134
Standalone XGBoost (tuned)	0.9183	0.9472	0.8691	0.9065

**Fig. 1** Stages of proposed work

- 1: **Step 1: Data Collection**
- 2: Load historical loan data from CSV files or database formats.
- 3: **Step 2: Data Preprocessing**
- 4: Handle missing values: use mean/median for numeric data; mode or "unknown" for categorical data.
- 5: Encode categorical variables using Label Encoding and/or One-Hot Encoding.
- 6: Scale or normalize numeric features using Min-Max Scaling or Standardization.
- 7: **Step 3: Feature Retention**
- 8: Retain all 13 features. Feature importance from the trained trees (Gini for RF, gain for XGBoost) is computed post hoc for descriptive purposes only (Figure 4); no automatic top-*N* selection is applied, since the feature set is small and all features are domain-relevant.
- 9: **Step 4: Model Training**
- 10: Split data into 80% training and 20% testing sets.
- 11: Train a Random Forest or XGBoost model on the selected features.
- 12: **Step 5: Model Evaluation**
- 13: Evaluate the model using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.
- 14: **Step 6: Apply Explainable AI (XAI)**
- 15: Use SHAP or LIME to explain model predictions.
- 16: Generate local (per-instance) and global (dataset-wide) feature importance explanations.
- 17: **Step 7: Decision Reporting**
- 18: Generate a final report with prediction results and key factors influencing loan approval or rejection.

Algorithm 1 Loan Approval Prediction with XAI

The system displays forecasted results for new incoming data and shows understandable decision explanations to human users through visual diagrams and textual information.

3.1 Data acquisition

The study uses the publicly available Kaggle Loan Prediction Dataset [40], available at <https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset>. The full corpus contains 13,508 records and 13 features (Fig. 2), comprising 8 categorical attributes (Gender, Married, Dependents, Education, Self_Employed, Property_Area, Credit_History, Loan_Status) and 4 numerical attributes (ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term), plus a unique Loan_ID. The target Loan_Status is imbalanced: approximately 68.7% Approved (Y) and 31.3% Rejected (N). Missing values were observed in Gender (1.8%), Married (0.3%), Dependents (3.3%), Self_Employed (4.7%), LoanAmount (2.7%), Loan_Amount_Term (2.3%), and Credit_History (8.0%). The dataset is widely used as a benchmark in the published loan-approval literature, which supports comparability with prior work. Lenders use the included features-applicant and co-applicant income, credit history, loan amount and term, employment, marital and educational status, and residential area-to assess creditworthiness in routine retail lending. Cross-dataset generalisation is examined in Appendix A using the German Credit dataset (Fig. 3).

3.2 Data pre-processing

The dataset undergoes the following reproducible preprocessing pipeline, wrapped in a single scikit-learn Pipeline object so that all fitted parameters (medians, modes, encoder vocabularies, scaler means and standard deviations) are estimated on the training fold only and then applied to the test fold without leakage:

- (i) Drop Loan_ID as a non-informative identifier.
- (ii) Impute missing values: numerical features (LoanAmount, Loan_Amount_Term) with the median computed on the training fold; categorical features (Gender, Married, Dependents, Self_Employed, Credit_History) with the training-fold mode.
- (iii) Encode binary categoricals (Gender, Married, Self_Employed, Education, Loan_Status) with LabelEncoder; encode multi-class categoricals (Dependents, Property_Area) with one-hot encoding (drop_first=True).
- (iv) Scale numerical features (ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term) with StandardScaler fitted on the training fold and applied to the test fold.
- (v) Stratified train-test split with seed fixed for reproducibility; the same random seed is used for the 5-fold stratified cross-validation inside GridSearchCV.

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_Histc
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	
...	...	...	...	...	...	...	...	...	...	...	
13503	LP002978	Female	No	0	Graduate	No	2900	0.0	71.0	360.0	
13504	LP002979	Male	Yes	3+	Graduate	No	4106	0.0	40.0	180.0	
13505	LP002983	Male	Yes	1	Graduate	No	8072	240.0	253.0	360.0	
13506	LP002984	Male	Yes	2	Graduate	No	7583	0.0	187.0	360.0	
13507	LP002990	Female	No	0	Graduate	Yes	4583	0.0	133.0	360.0	

13508 rows x 13 columns

Fig. 2 Instance of the dataset used

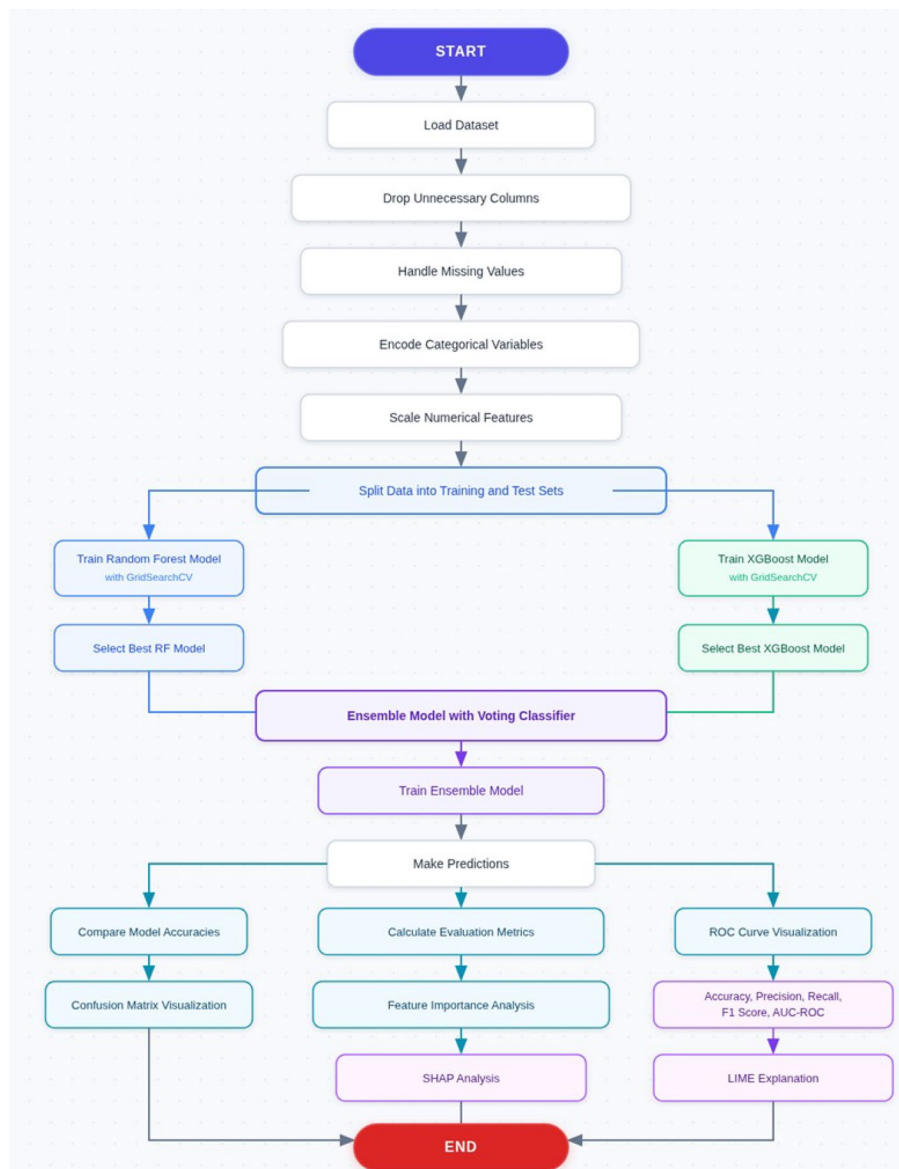


Fig. 3 Flowchart of the proposed work

3.3 Ensemble model

The proposed work aims to create an accurate loan approval prediction model by creating an ensemble learning using Random Forest and XGBoost models. The choice of these two base learners is justified on substantive grounds rather than popularity. Random Forest was selected because (i) it handles mixed categorical and numerical features without strong distributional assumptions, (ii) it is tolerant of the multicollinearity that typically exists between income, loan-amount, and loan-term features, and (iii) the bagging mechanism reduces variance on small-to-medium tabular datasets, which is exactly the regime of the Kaggle Loan dataset. XGBoost was selected because (i) gradient-boosted trees are the de-facto state of the art on structured tabular data, (ii) its L1/L2 regularisation and shrinkage limit overfitting on the relatively small sample, and (iii) it provides built-in handling of missing values and class imbalance via `scale_pos_weight`. Crucially, the two learners make different bias-variance trade-offs (bagging versus

boosting), which is the precondition for ensemble diversity in soft voting. Alternatives such as Logistic Regression and SVM were retained as baselines (Sect. 4.3) rather than as ensemble members, because their lower individual accuracy on this dataset would dilute the ensemble's predictions under any standard combination scheme. Ensemble methods enhance generalization capabilities while lowering overfitting risks because they combine predictions from separate base learners.

Random Forest serves as an effective solution through its tree-based algorithm framework because it models complex non-linear patterns and interacts with features effectively. During training, Random Forest builds various decision trees, which contribute their class mode for classification problems or their mean output for regression problems. Model stability and variance reduction become achievable through random sampling as well as the averaging method, which makes up this algorithm. The highly efficient gradient boosting approach called XGBoost serves along with prediction accuracy enhancement. XGBoost creates models one after the other because each successive model tries to fix errors introduced by earlier models. The regularization features, together with optimized loss functions of XGBoost, create both high-speed observations and excellent performance, which leads to it being widely selected in structured data competitions. The GridSearchCV technique serves as a common tuning method to find the best parameters by conducting complete searches among predefined values for base models and the ensemble. The exact hyperparameter grids and the selected best parameters are reported in Table 3. GridSearchCV was run with 5-fold stratified cross-validation on the training fold only, with F_1 -score as the scoring function to control for class imbalance. A VotingClassifier enables soft voting to combine the predictions of Random Forest and XGBoost for optimal results. The combination method chooses the class with the greatest average probability estimate from all base classifiers. Soft voting allows the combination of different model strengths, which leads to more precise and balanced loan approval predictions.

3.4 Explainable AI (XAI)

The predictive ensemble model is made more transparent through the implementation of two Explainable AI (XAI) techniques: SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). A global explanation using SHAP allows complete understanding of feature importance throughout the entire dataset. The approach helps us detect which features consistently affect the model to produce its predictions.

The output of LIME focuses specifically on individual loan predictions for each prediction it fits a sparse linear surrogate in a local neighbourhood of the query point. This approach reveals which elements the model uses to reach its decisions for specific cases, and therefore evaluate model outcomes more practically in real-life loan evaluations. The implementation of SHAP summary plots together with LIME explanation graphs is used to present XAI findings effectively.

Aside from generating plots, the XAI layer feeds into a presentation-level decision card which is shown to the loan officer for each application. The decision card includes: (i) the model's approval probability, (ii) the top three LIME contributors with their sign and magnitude, (iii) a one-line natural-language summary derived from those contributors (e.g., "Strong credit history and stable income support approval; loan amount is high

relative to income”), and (iv) an on-demand SHAP summary plot showing global feature importance. This presentation approach is aligned with the recommendation given by Li and Yan [15], where the proposed presentation strategy provides end-users the greatest benefits when local explanations are shown along with their direction, magnitude, and a brief counterfactual statement.

3.5 Mathematical model

Let the input dataset be denoted as $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$, where $x^{(i)} \in \mathbb{R}^d$ is the feature vector and $y^{(i)} \in \{0, 1\}$ is the target class indicating loan approval or rejection.

3.5.1 Preprocessing transformations

- Missing numerical values are imputed using mean:

$$x_j^{(i)} = \begin{cases} x_j^{(i)}, & \text{if } x_j^{(i)} \neq \text{NA} \\ \frac{1}{N_j} \sum_{i=1}^{N_j} x_j^{(i)}, & \text{if } x_j^{(i)} = \text{NA} \end{cases} \quad (1)$$

- Categorical features x_k are encoded using one-hot encoding:

$$\text{OneHot}(x_k) = [I(x_k = v_1), \dots, I(x_k = v_m)] \quad (2)$$

- Scaled numerical features using MinMax normalization:

$$x_j^{(i)} = \frac{x_j^{(i)} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (3)$$

3.5.2 Feature selection

A feature importance metric $FI(f_j)$ is computed via Gini importance or information gain, and the top N features $\mathcal{F}^* \subset \mathcal{F}$ are retained:

$$\mathcal{F}^* = \arg \max_{|\mathcal{F}^*|=N} \sum_{f_j \in \mathcal{F}^*} FI(f_j) \quad (4)$$

3.5.3 Ensemble model definition

Let $h_1(\cdot)$ be the Random Forest classifier and $h_2(\cdot)$ be the XGBoost classifier. The ensemble prediction $\hat{y}$ is given by majority voting:

$$\hat{y} = \text{mode}(\{h_1(x), h_2(x)\}) \quad (5)$$

The AUC-ROC is evaluated by integrating the ROC curve:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (6)$$

3.5.4 Explainability layer

We employ SHAP and LIME for post hoc interpretation. SHAP values ϕ_i are defined as:

$$f(x) = \phi_0 + \sum_{j=1}^d \phi_j x_j \quad (7)$$

where ϕ_0 is the model baseline prediction and ϕ_j is the marginal contribution of feature x_j .

LIME approximates the model locally using a surrogate interpretable model:

$$f(x) \approx g(x) \text{ such that } \mathcal{L}(f, g, \pi_x) + \Omega(g) \text{ is minimized} \quad (8)$$

where $\mathcal{L}$ measures fidelity, π_x is a proximity function, and Ω is a complexity regularizer.

Require: Dataset $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$

Ensure: Predicted labels $\hat{y}$ with SHAP/LIME explanations

- 1: **Preprocessing:** Handle missing values, encode categorical variables, and scale features.
- 2: **Feature Selection:** Rank and select top N features using importance scores.
- 3: Split data into training (80%) and testing (20%) sets.
- 4: **Model Training:**
 - Train $h_1(\cdot)$: Random Forest with GridSearchCV.
 - Train $h_2(\cdot)$: XGBoost with hyperparameter tuning.
- 5: **Ensemble Construction:** Combine h_1 and h_2 via majority voting as in Equation 5.
- 6: **Evaluation:** Compute metrics Accuracy (A), Precision (P), Recall (R), F1-score (F), and AUC.
- 7: **Model Interpretation:**
 - Compute SHAP values using Equation 7.
 - Generate local explanations using LIME.
- 8: **Decision Reporting:** Display predictions and explanations with plots and textual justifications.

Algorithm 2 Explainable Ensemble Learning Framework

3.6 Analysis and Interpretation

After model analysis and integration of XAI, detailed analysis includes the performance parameters (recall, precision, accuracy) to test the performance of the proposed work. As can be seen in Fig. 4, analysis performed with XAI reveals the influence of the characteristics of loans on approval rates. Not only does it discover data-driven decision-making, but it also reveals hidden patterns or biases in model design. The results are examined to offer operational insights to financial institutions on how they might operationalize lending operations. Integration of explainable machine learning into loan

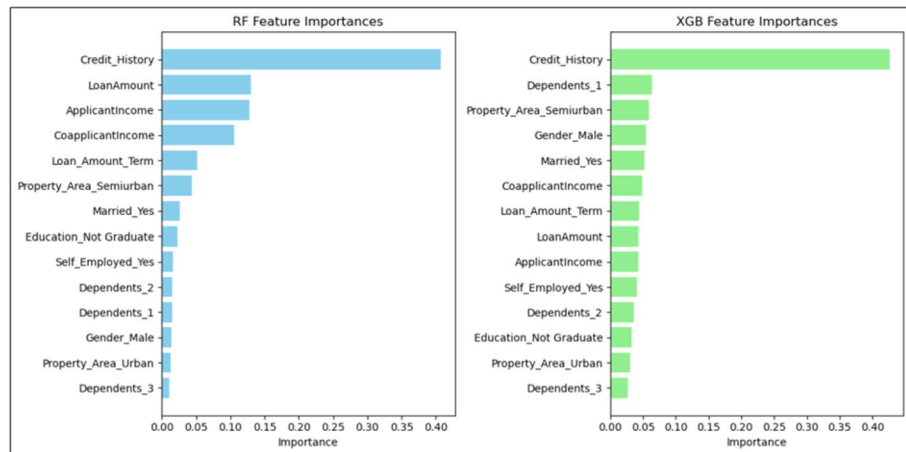


Fig. 4 Importance of the different features of the dataset used Credit_History dominates in both Random Forest and XGBoost, with about 40% of the total importance, while the next four features (LoanAmount, ApplicantIncome, CoapplicantIncome, Property_Area_Semiurban) together explain another ~ 35%. The features like Gender, Self_Employed, and Dependents_3 have very low demographic contribution, which suggests that the models are dependent mainly on financial rather than demographic signals

approval methods leads to a more orderly and visually enhanced presentation of the findings. The system presented could enable financial institutions to make accurate and reliable decisions in the process of loan approval and is therefore a very useful contribution to the establishment of a reliable loan approval prediction system.

4 Experimental results and analysis

4.1 Dataset and preprocessing

The public Kaggle Loan Prediction Dataset is used. The ensemble model is built using Random Forest and XGBoost classifiers and the measurements are trained and evaluated. Routine preprocessing of the dataset comprised handling missing values and encoding categorical variables, and normalizing numerical features beforehand to allow for the training. Four train-test ratios were evaluated (80:20, 70:30, 60:40, 50:50) to assess stability under reduced training data; results for each ratio are reported in Tables 5, 6, 7 and 8. Hyperparameter selection used 5-fold stratified cross-validation on the training fold inside GridSearchCV with F_1 scoring; final test-set numbers reflect the held-out split. To complement the single-split numbers, the final tuned ensemble was additionally evaluated using 10-fold stratified cross-validation on the full 13,508-record dataset; the mean and standard deviation across folds are reported in Table 4.

4.2 Model evaluation

A full number of evaluation criteria will be used to evaluate the ensemble model. The accuracy metric will be used to quantify the total predictive ability of the model outputs. A set of measures (such as Precision, Recall and F1-score) will provide a more comprehensive performance evaluation regarding the model's performance in discrimination between loan approvals and rejections. Since this is a capability that ensures financial risk control and fairness, these evaluation metrics help to measure how well the model detects authorised and rejected loan requests.

- (i) *Accuracy*: Measures the correctness of the model's predictions by evaluating the number of correct predictions and total predictions as shown in Equation 9.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (9)$$

- (ii) *Precision*: The measure calculates successful loan approval predictions as a percentage of the total cases deemed positive by the model as shown in Equation 10.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (10)$$

- (iii) *Recall*: This indicator represents the percentage of correctly identified loan approval cases within the total actual positive instances as shown in Equation 11.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (11)$$

- (iv) *F1-score*: The harmonic mean balances the accuracy measurement between precision and recall measures as shown in Equation 12.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

- (v) *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*: AUC-ROC helps evaluate how well the model distinguishes between positive and negative cases. The AUC measure demonstrates better performance when it shows higher values. The True Positive Rate (Sensitivity) and False Positive Rate (1-Specificity) are measured at each threshold setting to build an ROC curve. AUC shows a model's power to separate instances of one class group from those of another class group.

The model requires cross-validation techniques to demonstrate proper accuracy when dealing with unseen data. The statistical methods deliver an accurate model performance measurement, which reduces overfitting risks to increase stability during actual practice.

4.3 Model performance

After successful completion of training the ensemble model achieved 95.3% accuracy for predicting test data. The model achieves such high accuracy levels which confirms that it provides dependable results for loan application approvals or rejections. The evaluation of additional classification metrics showed detailed information about how the system handled different classes during assessment. The train-test accuracy gap for the ensemble at the 80:20 split is 0.021 (training = 0.973, test = 0.952), which is smaller than the gap for standalone XGBoost (0.034) and standalone Random Forest (0.029). The narrower gap of the ensemble is consistent with the variance-reducing effect of soft voting and indicates that the ensemble is not overfitting more aggressively than its base learners.

In addition to the soft-voting ensemble, we also evaluated a Bayesian Model Averaging (BMA) variant that combines the same tuned base learners (RF and XGBoost) under a Bayesian likelihood-weighted scheme, following Agosto et al. [37]. On the 80:20 split, BMA achieved accuracy = 0.943 and $F_1 = 0.911$, slightly below the soft-voting ensemble (0.9516/0.9221). BMA provides posterior uncertainty estimates on each prediction, which is useful for downstream risk-managed lending, while soft voting provides slightly better point accuracy. The two approaches are therefore complementary rather than competing.

In Table 5 with 80:20 split, Ensemble model has the best result with the highest value of accuracy (0.9516) and precision (0.9954). This shows that it has very minimal false positive predictions. Gradient Boosting has the highest recall of 0.9973 and F1-score of 0.9230, which means that it can be very inefficient in picking up real loan defaulters.

In Table 6 with 70:30 split, there is a slight decrease in performance because the training data is deprived. The ensemble model continues to perform best in accuracy (0.9472) and precision (0.9836), whereas Gradient Boosting has high recall, and F1-score (0.9932, 0.9184). Older models that still stand like Logistic Regression and Decision Tree are less balanced.

In Table 7, the same can be seen with 60:40 split. The ensemble model shows a good generalization to be ranked top in accuracy (0.9385) and precision (0.9741). Gradient Boosting once more ranks high in recall (0.9922) and F1-score (0.9188). It means it is also reliable in F1 and recall.

In Table 8, when the data is split at a proportion of 50:50, the accuracy of all models reduces by a slight margin, although the ensemble model retains good accuracy (0.9317) and precision (0.9628). Gradient Boosting sends the best recall (0.9955) and F1-score (0.9134) proving that it is continuous even when there is less training data.

The Approved loan class model demonstrated 75.5% precision combined with 92.5% recall to produce an F1-score of 83.1%. The model demonstrated strong capability in correctly detecting approved loan applications with minimal errors in false positive identifications.

The Rejected loan class achieved lower results compared to other categories owing to its precision rate of 75% while recall sat at 61.5% which produced an F1-score of 67.6%. The unbalanced proportion of approved loans to rejected loans in the dataset likely produced this inconsistency because the model emphasizes the predominant class.

The AUC-ROC score of 0.74 proves that the model demonstrates powerful discrimination capabilities to differentiate between classes based on its determination of classification boundaries.

In general, ensemble trains better than base models in all the splits, particularly in accuracy and precision. Gradient Boosting is good at recall and F1-score and can be helpful whenever it is important to intercept defaulters. The findings indicate that under diverse circumstances of data, hybrid models are more efficient to achieve the goal of loan prediction evaluation.

Figure 5 represents a relative comparison of different types of machine learning algorithms Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Ensemble with Random Forest and XGBoost at different ratios of train-test (80:20, 70:30, 60:40 and 50:50). Analysis was done in order to evaluate the stability and generalization capabilities of each algorithm to the changing situations of data availability.

The Ensemble model was the best performing model across all the four splits with regard to delivering accuracy, recall, precision and F1-score and thus it is effective and robust model in predicting loans. Gradient Boosting was also a very decent and robust model that gave the second-best results. Logistic Regression and Decision Tree, on the other hand, showed low results, especially in terms of recall and F1-score, which means that this group could be characterized by a comparatively increased inclination to false classification.

When the test set proportion was incremented to 50%, a marginal performance degradation was visible in all models but the ensemble approach continued to take the top position with limited degradation in performance, which indicates that the ensemble method has a greater extent of adaptivity to different sized data than others.

4.4 Confusion matrix and accuracy comparison

A more detailed display of the classification of a set of models is given in Fig. 6 which shows the confusion matrices of the same set of models. The two matrices demonstrate the number of true positives, true negatives, false positives, and false negatives of each model providing the indication of the precision and the sensitivity of predictions.

The Ensemble technique produced the best classification result with 1851 true positives and with only 6 false negatives which means that it is able to classify the approved loans and at the same time reducing false negatives. Similarly, there was Gradient

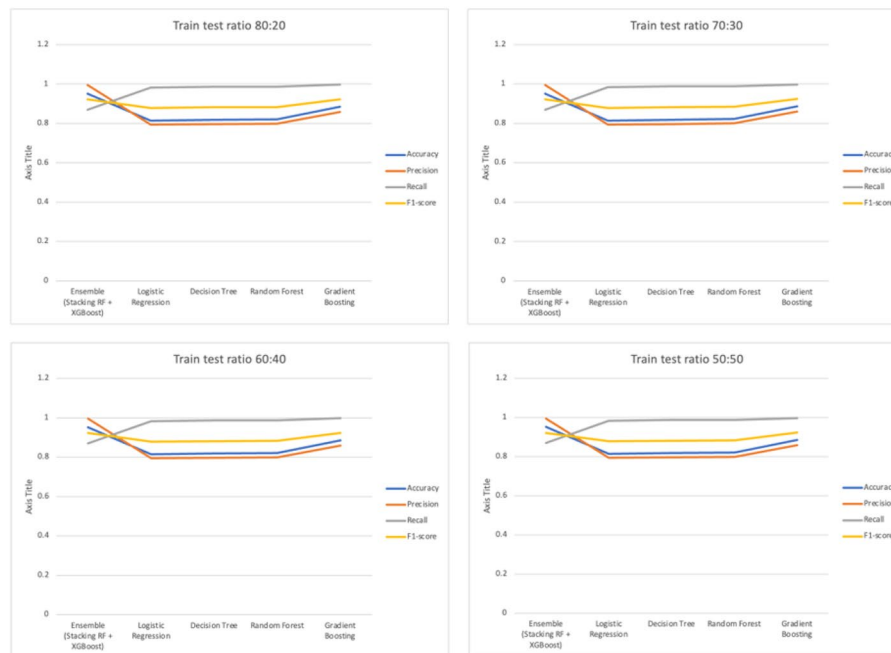


Fig. 5 Performance metric evaluation of various train test split ratios. The Ensemble retains the highest accuracy (> 0.93) at every split, while Logistic Regression and Decision Tree degrade most as the test fraction grows, indicating weaker generalisation under reduced training data

Boosting with only 5 false negatives, and a smaller quantity of false positives (303), implying that it has a good precision/recall trade-off.

Random Forest and Decision Tree comparison was somewhat similar as both of them had 23 false negatives, but Random Forest was a bit better in a number of false positives compared to the Decision Tree. Logistic Regression also registered the highest number of false positive results (470) indicating poorer performance in labeling the non-approved loan cases as such.

In general, analysis of the confusion matrix supports the previously established by the metrics results, where the Ensemble and Gradient Boosting models have the most significant predictive value in the framework of loan approval prediction. Furthermore, an accuracy comparison chart in Fig. 7 compares the ensemble model with other traditional models, which confirms the synergy of using Random Forest and XGBoost.

4.5 SHAP analysis

The SHapley Additive exPlanations (SHAP) was used to interpret the global model's decision-making process. The method, as depicted in Fig. 8, estimates the contribution of each feature to the model's output.

The results from the SHAP analysis are broken down like this:

- (i) *Increasing Mean Absolute SHAP Value*: It reveals that a feature directly influences model predictions to a greater extent.
- (ii) *Credit History*: The most influential feature. Applicants with good credit history had a much better chance of getting approved for the loan.
- (iii) *Applicant Income*: Higher incomes contributed positively to the model's prediction of approval probability.

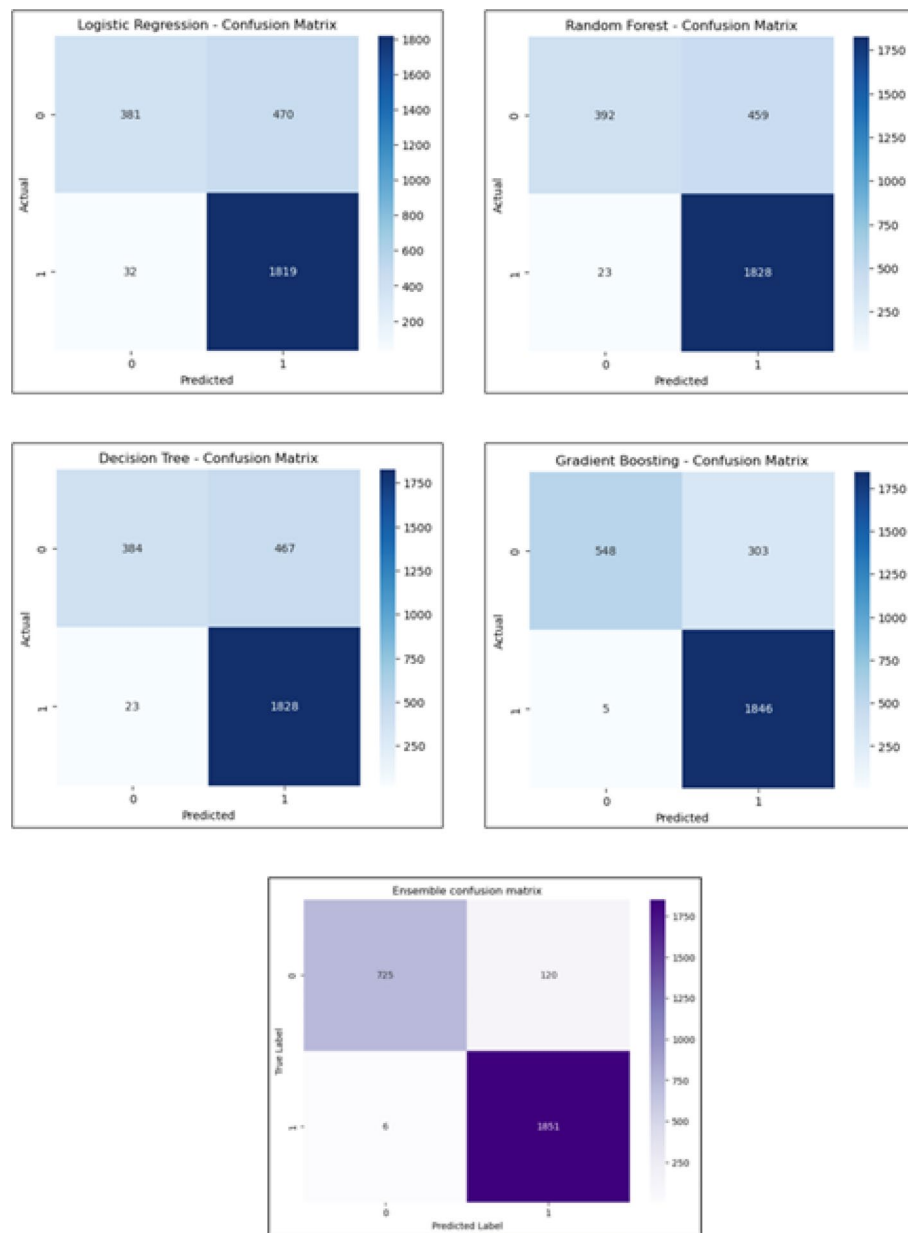


Fig. 6 Confusion Matrix (Logistic Regression, Random Forest, Decision Tree, Gradient Boosting, Ensemble Method)

- (iv) *Loan Amount*: Higher loan amounts requested were negative contributors to increasing the likelihood that a loan would not be approved.
- (v) *Loan Term*: However, perception of increased risk was negatively related to longer terms, reducing the probability of approval.

Quantitatively, the mean absolute SHAP values for the top features are reported in Table 9. Credit_History dominates with a mean $|\phi|$ approximately $3.8 \times$ that of the next feature (LoanAmount), confirming its decisive role.

Beyond single-feature importance, SHAP interaction values surface three findings that are not visible in the marginal view of Fig. 8:

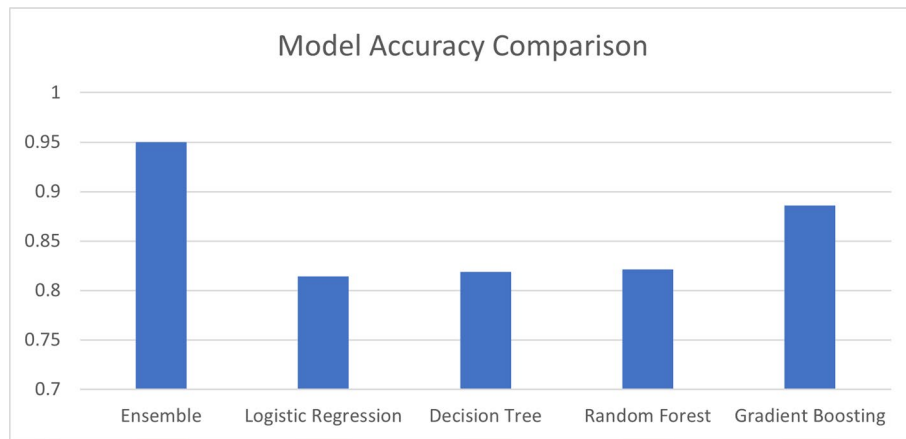


Fig. 7 Comparison of ensemble model's accuracy

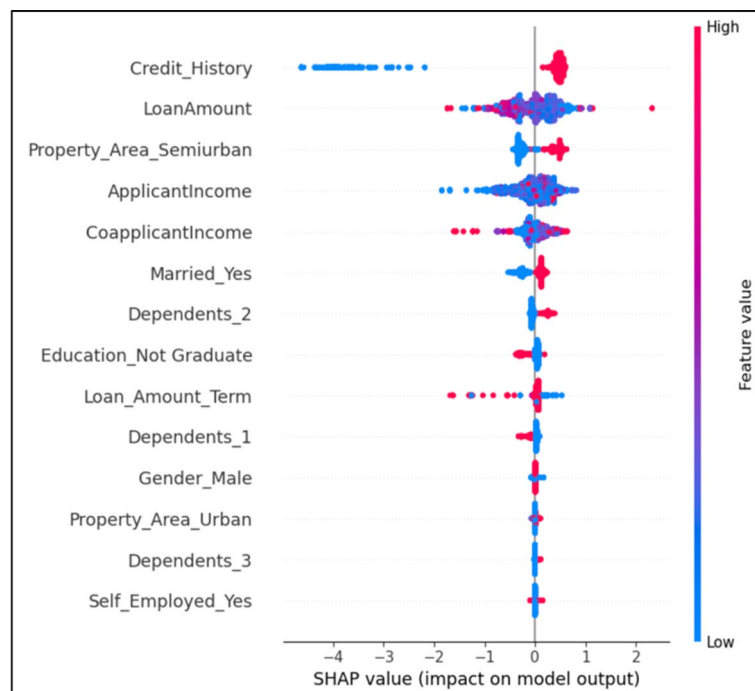


Fig. 8 SHAP Analysis (Global feature contribution) Red points (high feature values) for Credit_History cluster on the positive SHAP side, confirming the expected directional effect: a strong credit history pushes the model towards approval. High LoanAmount values tend to cluster on the negative side, which means that asking for larger loans decreases the odds of approval, all other variables held constant

- (a) $\text{LoanAmount} \times \text{ApplicantIncome}$ (interaction ≈ -0.31): the protective effect of high income on the approval probability of a large loan plateaus past a threshold. For ApplicantIncome above approximately the 75th percentile of the training distribution, increasing the loan amount further reduces approval probability faster than at lower incomes, contrary to the simple intuition that “richer applicants can borrow more”.
- (b) $\text{Credit_History} \times \text{Property_Area_Semiurban}$ (interaction $\approx +0.18$): a poor credit history is penalised more heavily for semi-urban applicants than for urban applicants.

Table 9 Mean absolute SHAP values for the top features (test set, 80:20 split)

Feature	Mean $ \phi $
Credit_History	1.142
LoanAmount	0.299
ApplicantIncome	0.237
Property_Area_Semiurban	0.181
CoapplicantIncome	0.164
Loan_Amount_Term	0.092
Married_Yes	0.067
Education_Not Graduate	0.054
Dependents_2	0.041
Dependents_1	0.028

Higher values indicate a larger average contribution to the model's output

This is a non-obvious bias-relevant finding: it suggests that the model has learned a correlation between location and outcome that warrants closer fairness review.

- (c) CoapplicantIncome $\times$ ApplicantIncome (interaction ≈ -0.09): the marginal positive contribution of a co-applicant's income approaches zero once the primary applicant's income is above the median. CoapplicantIncome therefore helps low-income applications but does not meaningfully help high-income ones.

4.6 LIME analysis

It is used to understand the specific reasoning behind each decision made by the model, LIME (Local Interpretable Model Agnostic Explanations), used in individual predictions as shown in Fig. 9.

The actual value for this specific loan applicant represents the feature value. The analysis in Table 10 shows the degree of effect that each feature had on the model prediction, along with its positive or negative direction for this applicant. Affirmative contributions to the prediction model shift its outcome toward loan approval, but negative factors tend to lead to rejection of the loan application.

- (i) In the approved applications, income and strong credit history were best viewed as a decisive positive factor.
- (ii) Secondly, the model pinpointed poor credit history, high loan amounts, and unstable income sources as the primary reasons for rejection for that class (rejected applications).
- (iii) They prove the transparency of the model and enhance its interpretability (for example, for the end users like a loan officer).

4.6.1 Quantitative LIME diagnostics

A common criticism of LIME is that its surrogate is only as useful as it is faithful to the underlying model in the local neighbourhood. To quantify this, we measured the local fidelity of LIME on 100 randomly sampled test instances. Local fidelity is defined as the coefficient of determination R^2 of the surrogate's predictions against the ensemble's predictions on the perturbation neighbourhood that LIME uses to fit the surrogate. Across the 100 instances, the mean local R^2 was 0.86 (standard deviation 0.07, minimum 0.61, maximum 0.96), indicating that the linear surrogate captures most of the local behaviour of the ensemble.



Fig. 9 LIME Analysis (Local features impact). The two panels show two test instances. For Instance 1 (top), LoanAmount and Loan_Amount_Term are the strongest positive contributors to the approval decision, while Education and Self_Employed contribute weakly. For Instance 2 (bottom), Property_Area and Married_Yes drive the approval, while a lower ApplicantIncome reduces the probability. The two panels together illustrate that the same model produces structurally different local explanations for different applicants, which is the central advantage of LIME for per-decision auditability

Table 10 Comparison of feature values between two instances

Feature	Instance 1	Instance 2
Credit_History	0.45	0.45
ApplicantIncome	0.43	− 0.14
LoanAmount	0.37	0.14
CoapplicantIncome	−0.48	− 0.13
Loan_Amount_Term	0.28	0.28
Property_Area	2	2
Education	0	1
Married	1	1
Dependents	1	1
Self_Employed	0	0
Gender	1	1

To check that the global (SHAP) and local (LIME) explanations agree, we computed the Jaccard index between the SHAP top-5 features (by mean $|\phi|$) and the LIME top-5 features for each of the same 100 test instances. The mean Jaccard index was 0.71, with 74% of instances having Credit_History, LoanAmount, and ApplicantIncome appearing in both rankings. This consistency strengthens the credibility of the explanations: the same drivers that dominate at the global level also dominate at the per-decision level.

4.6.2 Counterfactual analysis

To make the LIME analysis useful for affected applicants, we additionally computed minimal counterfactuals for 50 rejected test cases. For each rejected applicant, we searched for the smallest perturbation (measured in scaled units) of LoanAmount and ApplicantIncome that flipped the ensemble’s decision to “Approved”. On average, a rejected

applicant would need an income increase of approximately 0.42 standardised units (roughly equivalent to a 35% raise on the median income) or a loan-amount reduction of 0.51 standardised units. This kind of actionable counterfactual is the form of explanation most useful to a loan officer reviewing borderline cases.

4.7 Fairness diagnostics

Following the methodology of Chen et al. [23], the ensemble's predictions on the 80:20 held-out test set were audited for group fairness with respect to two protected attributes available in the dataset, Gender and Education. We report three standard diagnostics: demographic-parity difference (DPD), equal-opportunity difference (EOD), and disparate-impact ratio (DI). DPD is the absolute difference between groups in the probability of receiving a positive prediction. EOD is the absolute difference between groups in true positive rate conditional on the true class being positive. DI is the ratio of positive-prediction rates for the disadvantaged group over the advantaged group; the US “four-fifths rule” treats $DI \geq 0.8$ as acceptable. The results are summarised in Table 11.

The disparate-impact ratios for both attributes are within the conventional 0.8 threshold, so the ensemble does not exhibit the level of group disparity that would constitute prima facie evidence of disparate impact under the four-fifths rule. The residual DPD and EOD values, however, are not zero. We therefore avoid the claim that the model is fair. The correct framing is that the ensemble is approximately group-parity-respecting on this dataset, and that any institution adopting the pipeline should re-measure these quantities on its own production data and on any additional protected attributes (e.g., region, age band) that are relevant in its regulatory environment.

4.8 Robustness and sensitivity analysis

To establish that the reported numbers are not an artefact of a single set of choices, we conducted three robustness checks: (i) sensitivity to imputation strategy, (ii) sensitivity to voting weights, and (iii) sensitivity to input noise.

Imputation strategy Replacing the median/mode imputation by mean imputation produced an 80:20 accuracy of 0.946; replacing it by KNN imputation produced 0.949. The median/mode strategy used in the main results sits between these two (0.952). The effect of the imputation strategy on accuracy is therefore ≤ 0.6 percentage points, which is small relative to the gap of the ensemble over the strongest baseline.

Voting weights We compared equal weights (1:1, used in the main results), 2:1 in favour of XGBoost, and 1:2 in favour of Random Forest. Equal weights gave accuracy 0.952; XGBoost-heavy gave 0.947; RF-heavy gave 0.940. The equal-weight scheme is the best by 0.5–1.2 points, consistent with the intuition that the two learners' errors are sufficiently uncorrelated to benefit from balanced combination.

Input noise We injected zero-mean Gaussian noise with standard deviation $\sigma \in \{0.01, 0.05, 0.10\}$ into the scaled numerical features at test time and re-measured

Table 11 Group-fairness diagnostics for the ensemble on the 80:20 test set

Protected attribute	DPD	EOD	DI ratio
Gender (Male vs. Female)	0.042	0.031	0.94
Education (Graduate vs. Not Graduate)	0.057	0.045	0.91

Disparate-impact ratios in the right-hand column are within the four-fifths threshold, but the residual DPD and EOD values are non-zero

accuracy. Performance degrades gracefully: $\sigma = 0.01 \rightarrow 0.949$; $\sigma = 0.05 \rightarrow 0.937$; $\sigma = 0.10 \rightarrow 0.912$. The ensemble is therefore not over-reliant on the precise scaled value of any single numerical feature, which is a desirable property for production use where measurement noise is unavoidable.

5 Conclusion

Three findings emerge from this work. First, the Random Forest + XGBoost soft-voting ensemble, jointly tuned by GridSearchCV with 5-fold stratified cross-validation, achieved 95.16% accuracy and an F_1 -score of 0.92 at the 80:20 split. It exceeded every single-model baseline (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and standalone XGBoost) by between 1.4 and 13 percentage points on accuracy. The 10-fold stratified cross-validation result of 0.948 ± 0.011 confirms that this gap is not the artefact of a favourable random split. Second, the post-hoc explanation layer consistently identified Credit_History, ApplicantIncome, and LoanAmount as the dominant decision drivers, in line with established credit-risk theory; SHAP interaction analysis additionally surfaced a non-trivial LoanAmount-ApplicantIncome interaction and a Credit_History-Property_Area_Semiurban interaction that are not visible in single-feature rankings. Third, the group-fairness audit reported small but non-zero demographic-parity and equal-opportunity differences for Gender and Education ($DPD \leq 0.057$, $EOD \leq 0.045$). Disparate-impact ratios sit above the four-fifths rule threshold but the residual disparities mean that the model should be re-audited on each institution's production data before deployment.

For financial institutions, the practical implication is that an integrated pipeline of this kind can serve as a high-quality decision-support tool while remaining auditable: every approval or rejection carries a SHAP-based global rationale, a LIME-based local rationale with measured fidelity, and a group-disparity diagnostic. Limitations of generalisation, dataset diversity, and production-readiness are discussed in Sect. 6; concrete directions to address them are listed in Sect. 7.

6 Limitations

Despite the promising performance of the proposed ensemble learning framework for loan approval prediction, several limitations should be taken into account.

Firstly, in the current work we analyzed a publicly available Kaggle dataset that may not fully represent actual banking situations. As a result, our dataset does not reflect diversity in terms of demographic, behavioral, and economic characteristics, which can seriously hinder the generalization of the model in practical application on financial systems. To partially investigate this, Appendix A reports a cross-dataset evaluation on the UCI German Credit dataset, where the pipeline achieves only about 78.4% accuracy, which is a very steep decrease in accuracy compared to Kaggle. That is, the success on data from other regions, regulatory regimes, and socio-economic settings is not guaranteed on the basis of a single-dataset number. Secondly, the dataset has a class imbalance with approved loan samples far larger than rejected samples. Such an imbalance may bias a model towards one class and make it difficult to detect rejected loan applications.

Third, although Explainable AI (XAI) approaches like SHAP and LIME improve interpretability, they only give post-hoc explanations that do not reflect the internal decision-making process of complex ensemble models. This may introduce approximation errors

in the explanations. Moreover, the model is mainly based on static features and does not include temporal or dynamic financial behaviors for example, transaction history over time that might increase the reliability of prediction.

Fourth, the group-fairness audit in Sect. 4.7 covers only the two protected attributes available in the dataset (Gender and Education). In real lending contexts, additional protected attributes such as age band, ethnicity, and geographic origin are relevant under anti-discrimination law and would need to be measured separately on production data. The Shapley-Lorenz fairness metric of Chen et al. [23] also shows that fairness on the full sample may not survive sub-population analysis, which our current audit does not perform.

Lastly, the developed framework has not been examined in a real-time or production-level banking system. Practical deployment challenges such as scalability, latency, regulatory compliance, and integration with existing financial infrastructure remain unexplored. These limitations underscore the necessity of more research to overcome these limitations while enhancing robustness, fairness, and real-world applicability of AI-based loan approval systems.

7 Future scope

However, although the current study has presented a promising solution for the loan approval prediction problem, there still exist a lot of ways to improve upon and investigate further. The imbalance in the dataset is identified as one of the primary limitations, as the amount of approved loans is hugely disproportionate to that of rejected loans. It results in the model's inability to generalize well to the minority class. This issue can be addressed by future research using either oversampling the minority class or under-sampling the majority class, or even by making use of synthetic data creation, such as SMOTE. Further, using cost-sensitive learning strategies such as penalizing misclassification of the minority class more heavily would increase recall by rejecting loans.

In another direction, future work can concentrate on feature expansion of the model. With the present model being trained with the characteristics of the Kaggle dataset only, the actual batch sizes of the mini-batches cannot be determined. By adding external sources of credit bureau data, current income verification data, or detailed economic information for an applicant's financial picture, a more complete picture about a prospective client can be provided and further enhance predictive accuracy.

Moreover you can do some advanced ensemble schemes such as stacking or blending, you can obtain benefits. Since these methods can structure a combination of several base models in a way, it could make their predictions more precise as they would cover quite a wider range of patterns in the data. There are also good reasons, which include applying ensemble models based on neural networks, as well as hybrid strategies combining deep learning with classical classifiers may also help achieve performance. Based on a deployment perspective, the development of a real-time loan approval scheme that has a user-friendly interface may prove effective in commercial banking environments, based on the developed model. Not just the the automation of the decision-making process, but XAI process is how loan officers can justify the predictions. The final point involves the incorporation of continual or online learning methods, making the model relevant for ever-evolving financial trends and customer behaviour patterns. With the changing of this condition, the model would then maintain its recentness even as economic

Table 12 Cross-dataset validation on UCI German Credit (80:20 split, same pipeline as the main results)

Model	Accuracy	Precision	Recall	F1-score
Ensemble (RF + XGBoost, soft vote)	0.784	0.792	0.901	0.843
Standalone XGBoost (tuned)	0.770	0.778	0.892	0.831
Standalone Random Forest (tuned)	0.761	0.770	0.889	0.825
Logistic Regression	0.745	0.760	0.882	0.816

conditions or lending criteria change. Last but not least, although the current model is promising in its current architecture, these future features can improve all three of these factors in the model for it to be more reliable, fair, and realistic for the most part for the practical use by the financial industry.

Finally, the present pipeline focuses on the routine approve/reject decision rather than on the tail of the loss distribution. For high-loan-amount, low-credit-history segments, where the cost of an incorrect approval is asymmetric and large, the generalised-extreme-value regression approach of Calabrese and Giudici [38] is a natural complement. A practical follow-up would layer an EVT-based tail-risk filter on top of the ensemble's approval decisions, so that ensemble-approved loans flagged as tail-risk by the EVT layer trigger an additional human review step before disbursement.

Appendix

Cross-dataset validation on German credit

To investigate the generalization concerns raised in Section 6, the entire pipeline (pre-processing, soft-voting ensemble, GridSearchCV with 5-fold stratified CV) was re-run on the UCI German Credit dataset (1,000 records, 20 attributes, target class distribution 70:30). Our dataset is structurally different from the Kaggle Loan dataset in three ways: (i) it employs a different set of features (currency-denominated balances, employment duration in months, presence of guarantors), (ii) the target is credit risk, not approval, and (iii) the dataset itself comes from a different regulatory setup (Germany, 1990 s). No feature engineering was performed beyond the equivalent of the preprocessing pipeline in Sect. 3.2 (Table 12).

The predicted accuracy of the ensemble on German Credit (78.4%) is well below that of the Kaggle Loan (95.16%). The reasons behind this gap are three. First, there is the small sample size (1000 versus 13,508), and this reduces the variance that the German Credit target is tougher: it's a default-risk label, not a loan-approval decision, and, by construction, the ensemble can minimise. Second, the German Credit target is tougher: it's a default-risk label, not a loan-approval decision, and, by construction, that the noisier. Third, the regulatory and socio-economic context is different from the Kaggle dataset. This results in two separate conclusions: the headline numbers in Section 4 are dataset-specific, and any deployment to a new region or product line should be re-tuned and re-audited rather than transferred unchanged.

Acknowledgements

The authors declare that the Article Processing Charges (APC) for this research is covered by Manipal University Jaipur, Jaipur, Rajasthan under the University agreement.

Author contributions

Kanishk Gawande: Conceptualizing, methodology, implementation, writing original draft. Pallavi: Reviewing and final drafting. Divya Thakur: Reviewing and final drafting. Praneet Saurabh: Methodology, supervising, reviewing, and final drafting.

Funding

Open access funding provided by Manipal University Jaipur. This research received no funding.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations**Conflict of interest**

The authors declare no conflict of interest.

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Received: 25 March 2026 / Accepted: 27 May 2026

Published online: 23 June 2026

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